AUDITORNE VJEŽBE 7.

Konvolucija

Zadatak 1.) Odziv disk. sustava na pobudu $u[n]=\alpha^n \mu[n]$, ako je poznati impulsni odziv sustava: $h[n]=\beta^n \mu[n]$.

Konvolucijska sumacija: $y[n]=\sum_{m=-\infty}^{\infty} u[m]\cdot h[n-m] = (u*h)[n]$

$k=n-m$

$$= \sum_{k=-\infty}^{\infty} h[k]\cdot u[n-k] = (h*u)[n]$$

$y[n]=?$

$$y[n]=\sum_{m=-\infty}^{\infty} h[m]\cdot u[n-m]$$

geom. red

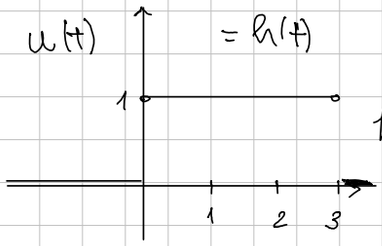
$$y[n]=\sum_{m=-\infty}^{\infty} \beta^m \mu[m]\cdot \alpha^{n-m}\mu[n-m] = \sum_{m=0}^{n} \beta^m\cdot\alpha^n\cdot\alpha^{-m} = \alpha^n \sum_{m=0}^{n}\left(\frac{\beta}{\alpha}\right)^m$$

$$y[n]=\alpha^n \frac{1-\left(\frac{\beta}{\alpha}\right)^{n+1}}{1-\frac{\beta}{\alpha}} = \frac{\alpha^n-\frac{\beta^{n+1}}{\alpha^1}}{1-\frac{\beta}{\alpha}} = \frac{\alpha^n-\beta^{n+1}}{\alpha-\beta}$$

Zadatak 2.) Nađite odziv kontinuiranog sustava na pobudu

$u(t)=\begin{cases}1, & 0<t<3\\ 0, & \text{inače}\end{cases}$ ako je impulsni odziv $h(t)=\begin{cases}1, & 0<t<2\\ 0, & \text{inače}\end{cases}$

$y(t)=\sum_{m=-\infty}^{\infty} u[m]\,h[n-m]$ odnosno $y(t)=\int_{-\infty}^{\infty} u[\tau]\cdot h(t-\tau)\,d\tau$



$u(t)=\mu(t)-\mu(t-3)$

$h(t)=\mu(t)-\mu(t-2)$

$$y(t)=\int_{-\infty}^{\infty}(\mu(\tau)-\mu(\tau-3))\cdot(\mu(t-\tau)-\mu(t-2-\tau))\,d\tau$$

$$y(t)=\int_{-\infty}^{\infty}\mu(\tau)\cdot\mu(t-\tau)-\mu(\tau)\mu(t-2-\tau)-\mu(\tau-3)\mu(t-\tau)+\mu(\tau-3)\mu(t-2-\tau)\,d\tau$$

$\mu(\tau)\cdot\mu(t-\tau)\begin{cases}1, & 0\le\tau\le t\\ 0, & \text{inače}\end{cases} \rightarrow \mu(t)\int_0^t d\tau$ ✓

$\mu(\tau)\cdot\mu(t-2-\tau)\begin{cases}1, & 0\le\tau\le t-2\\ 0, & \text{inače}\end{cases} \Rightarrow \mu(t-2)\int_0^{t-2} d\tau$ ✓

$\mu(\tau-3)\cdot\mu(t-\tau)\begin{cases}1, & 3\le\tau\le t\\ 0, & \text{inače}\end{cases} \Rightarrow \mu(t-3)\int_3^t d\tau$

$\mu(\tau-3)\cdot\mu(t-2-\tau)\begin{cases}1, & 3\le\tau\le t-2\\ 0, & \text{inače}\end{cases} \Rightarrow \mu(t-5)\int_3^{t-2} d\tau$

$$y(t)=\mu(t)\int_0^t d\tau-\mu(t-2)\int_0^{t-2} d\tau-\mu(t-3)\int_3^t d\tau+\mu(t-5)\int_3^{t-2} d\tau$$

$$\boxed{y(t)=\mu(t)\cdot t-\mu(t-2)\cdot(t-2)-\mu(t-3)\cdot(t-3)+\mu(t-5)\cdot(t-5)}$$

Zadatak 3.) Dokažite svojstva konvolucije vremenski kont. sustava

a) $u(t) * \delta(t) = u(t)$

$$\int_{-\infty}^{\infty} u(\tau)\cdot\delta(\underbrace{t}_{t_0}-\tau)\,d\tau = u(t)\underbrace{\int_{-\infty}^{\infty}\delta(t-\tau)\,d\tau}_{1} = u(t)\cdot 1 = u(t)$$

b) $u(t) * \delta(t-t_0) = u(t-t_0)$

$$\int_{-\infty}^{\infty} u(\tau)\cdot\delta(\underbrace{t-t_0}_{t_0'}-\tau)\,d\tau = u(t-t_0)\int_{-\infty}^{\infty}\delta(t-t_0-\tau)\,d\tau = u(t-t_0)\cdot 1 = u(t-t_0)$$

c) $u(t) * \mu(t) = \int_{-\infty}^{t} u(\tau)\,d\tau$

$$\int_{-\infty}^{\infty} u(\tau) * \mu(\underbrace{t}_{\text{gornja granica}}-\tau)\,d\tau = \int_{-\infty}^{t} u(\tau)\,d\tau$$

d) $u(t) * \mu(t-t_0) = \int_{-\infty}^{t-t_0} u(\tau)\,d\tau$

$$\int_{-\infty}^{\infty} u(\tau)\,\mu(\underbrace{t-t_0}_{\text{gornja granica}}-\tau)\,d\tau = \int_{-\infty}^{t-t_0} u(\tau)\,d\tau$$